BGIN Block#11 Meeting Report

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Institutional Adoption and Regulatory Frontier for Emerging Technologies

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BGIN Block#11

Institutional Adoption and Regulatory Frontier for Emerging Technologies

Date: Monday, October 21, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Session Objectives:

- **Focus Areas:** Address gaps in crypto regulation by discussing less-explored regulatory topics and hot-button issues without regulatory clarity.
- **Emerging Tech & Institutional Adoption:** Highlight challenges associated with new technologies and institutional crypto use.
- **Outcome Goal:** Develop a document listing regulatory uncertainties and unidentified risks that may require future regulatory guidance.

U.S. Regulatory Overview (Jason Brett):

- **Background:** Jason Brett, former FDIC regulator, emphasized the slow nature of U.S. legislative processes in crypto, particularly in stablecoin regulation.
- **Federal vs. State Regulatory Tensions:** Key issue for stablecoin regulation lies in whether federal entities like the FDIC or the Federal Reserve or state-level agencies will have regulatory authority, leading to potential conflicts.
- **Stablecoin Significance:** Stablecoins are seen as vital due to their role as U.S. dollar derivatives with full backing, making them relevant to both monetary stability and consumer protection.
- **Potential Legislative Timeline:** Anticipated movement on stablecoin regulation in a post-election lame-duck Congress session, as stablecoins could be part of “must-pass” legislation.

Significant Events Shaping U.S. Crypto Policy:

- **Facebook Libra's Impact (2019):** Raised awareness among U.S. lawmakers about the potential of global stablecoins, sparking early regulatory discussions.
- **2021 Crypto Tax Proposal:** Catalyzed the crypto industry’s lobbying activities, increasing advocacy and regulatory engagement within Washington, D.C.

- **Crypto's Role in 2024 Election:** Rising public interest in crypto policy positions of candidates, reflecting voter sentiment and showing crypto's importance as an election issue.

Global Regulatory Perspective (Josh Besco):

- **Role at Talos:** Josh Besco, head of compliance strategy, discusses Talos's role as an infrastructure provider connecting institutions to crypto markets while navigating global regulations.
- **Global Regulatory Comparisons:** Notes other countries, such as those in Europe with MiCA, are progressing faster with comprehensive crypto regulation.
- **Single-Issue Crypto Voters:** The U.S. has a surprising number of voters for whom crypto is the main voting issue, highlighting crypto's unique appeal.
- **U.S. Industry's Desire for Regulation:** Contrary to public perception, the crypto industry is eager for regulatory clarity to foster stability and legitimacy.

Institutional Adoption (Leslie Chapkin, Paxos):

- **Bank Partnerships Essential:** Crypto firms still need banking partners for services like asset safeguarding, compliance, and efficient transactions.
- **Regulatory Deterrents:** Current regulatory guidance discourages banks from engaging with crypto firms, impacting their operations.
- **Consequences of Bank Closures:** The closure of crypto-friendly banks (e.g., Silvergate, Signature) forces firms to rely on smaller banks, raising concentration risk.
- **Stablecoin Custody Advocacy:** Urges U.S. regulators to permit domestic banks to offer stablecoin custody services to support financial stability.

Barriers to Institutional Crypto Adoption:

- **Regulatory Constraints (SAB 121):** Restrictions prevent banks from entering the crypto custody space, complicating institutional investment.
- **Institutional Diversification Needs:** Asset managers seek to diversify portfolios with crypto but require dependable custody options, currently limited by regulatory barriers.
- **Decentralization vs. Intermediaries:** Emerging decentralized technologies often lack traditional intermediaries, challenging existing regulatory models that rely on intermediaries as control points.

Current Regulatory Sentiment & "Operation Chokepoint 2.0":

- **Regulatory Guidance Against Crypto:** U.S. regulators have adopted a restrictive stance, overtly discouraging banks from crypto involvement, impacting banking support for crypto firms.

- **Growing Interest Among Agencies:** Despite the restrictive tone, there's a significant push within agencies to understand crypto and develop sensible frameworks.

International Regulatory Efforts & Harmonization:

- **Global Regulatory Landscape:** Countries like the UAE, Singapore, Japan, and the EU have created more comprehensive crypto regulations, leading to a fragmented global approach.
- **Potential U.S. Leadership Role:** The U.S. could unify international standards, especially for stablecoins and cross-border regulatory issues, promoting a cohesive regulatory environment.

Institutional Demand for Regulatory Clarity and Custody Solutions:

- **Custodial Needs for Crypto Investments:** Traditional asset managers rely on third-party custodians for security and regulatory compliance, which are currently lacking in crypto.
- **SAB 121 and Custody Restrictions:** Clarity around crypto custody requirements, particularly under SAB 121, would unlock significant institutional capital for crypto markets.

Crypto Custody & Security Standards:

- **Proof-of-Reserve Requirements:** Implementing proof-of-reserve standards would improve transparency and trust in stablecoins and crypto assets.
- **Stablecoin Industry Transparency:** A U.S.-based, regulated stablecoin industry would help eliminate opacity and address security concerns, improving overall stability.

Technology-Specific Regulatory Challenges:

- **Lightning Network:** Requires an understanding of its decentralized structure, self-custodial wallets, and intermediary nodes.
- **Emerging Technologies (DeFi, Liquid Staking):** Layer 2 technologies, cross-chain protocols, and DeFi applications introduce operational, AML, and security risks that regulators need to assess.

Need for Common Terminology and Frameworks:

- **Unified Terminology:** Consistent language for describing decentralized and centralized elements of crypto could ease regulatory understanding and support industry growth.

- **Visual Tools for Regulators:** Flow diagrams, analogies, and structured visuals can clarify crypto complexities, aiding regulators in comprehending new tech without heavy jargon.

Conclusion:

- **Proactive Regulatory Frameworks Needed:** Regulations should balance innovation with safety by acknowledging crypto's unique characteristics.
- **Collaboration with International Regulators:** Unified global standards, especially around custody, cybersecurity, and risk management, would streamline compliance and support industry growth.

Transcript

This session is to discuss what we haven't discussed a lot in general and the current registration. So we can discuss, we are trying, so we want to discuss like hot discussion topics, but you know we haven't had regulatory clarification yet. So while the topic is institutional adoption, we would like to talk about emerging technology, the challenges yet as well. But the ideal goal of this section is to create a list of regulatory frontier regulatory uncertainties to analyze the risks regulators wanted to see. And we are thinking about creating documents in the beginning about the risk we haven't actually identified yet.

Yeah, we can also discuss the details of some of them, including institutional adoptions, of course. But we would like to start from talking about US regulation because now that we are in the US, not everybody is probably familiar with US regulation because people come from many different places. So I would like to kick off by asking Jason, what are the directions of regulation considering current submitted and past bills in the US and also considering the US election and the industry trend?

Sure, yeah. Thanks very much. My name is Jason Brett. I'm a former FDIC regulator during the global financial crisis, the FDIC 2008–2011, and I did some consulting with the Treasury, the home portable modification program as an appliance examiner for years. Since 2016, I've been in crypto. If it was on, is this work? Okay. Thanks. Yeah. Always good to have a microphone on. Yeah, so this former US regulator, and then in 2016, I was the director of policy for the Chamber of Digital Commerce for about a year. Then I helped set up the government affairs shop here in DC for Ethereum and Consensus for Derlumen.

2018 and 2019, since then I just have my own consultancy where I help various blockchain companies and others set up their government affairs office for here in the US or, you know, establish lobbying concepts. You know, we talk about different bills and where the US is. Like, you know, one thing you should know about the United States: it takes a long, long time for a bill to come to fruition. The Stablecoin bill has been in the works now for almost four or five

years. I helped actually co-author the first stablecoin bill that didn't go very well because they said their banks could use it. And that was not my plan originally. But where we stand now is stablecoins are very close.

The big question has to do with who's going to regulate the stablecoins. Is it going to be at the federal level? Was it the FDIC, like the Federal Reserve would be involved, or is it going to be at the state level? You can hear from Lesley and Patric, who say they are very highly regulated already by the New York Department of Financial Services. And I think one of the big blockers of that bill, to be honest, has been the state-to-federal tension of who's going to actually be able to regulate stablecoins. And I think that's with good measure because stablecoins are representing the US dollar. You know, when people are using them, they're essentially using what is a derivative of the US dollar that is backed 100%. So a lot of times people will tell you about the banking crisis in 2023 and talk about how maybe somehow stablecoins had something to do with it or crypto had something to do with it. It actually more has to do with how the bank was managed and how the regulators were looking at those banks. And it has to do with, obviously, the fragmented banking system where we have just a portion of that banking system that supports us versus like 100% backed. So stablecoins are 100% backed by the dollar. It started when the OCC allowed it in 2020, with allowing bank regulations for banks to hold deposits as stablecoins.

So there's lots of questions about it. The other one is, of course, the peg. How do you manage the peg of the stablecoin? A lot of times people thought about if maybe different banks started creating their own stablecoins. Are they going to hold the peg, or would they maybe strategically drop the peg to try to increase the number of customers they have? Companies like Paxos and Circle and others have been fighting this fight for a long number of years. And while there have been some tests of the peg of stablecoins, they still have been backed 100%. It's the fastest, cheapest way of moving money around the world in a very secure environment. And it's an improvement on the current banking system. So I think you might see some stablecoin legislation at the end of this year, even though we'll have a lame-duck Congress, it'll be after the election.

So by then we'll know if either Trump or Harris is going to be the new incoming president. And lame-duck Congresses can provide this type of atmosphere for possible legislation. So one of the things is it's probably going to be discussed with what's called a must-pass piece of legislation, which is when you have a piece that has to pass because it's like the budget or the defense side. And they try to figure out how stablecoins might fit in that. So it's really important right now for everyone in the industry to be working with the leaders in Congress, like Chuck Schumer and others, because they're going to be at that negotiating table figuring out what that stablecoin bill may or may not look like.

We heard a big development in stablecoins, and I'll leave it at this: Maxine Waters and the House Financial Services Committee, one of the last hearings, sort of reached across the aisle to the Republicans and said she wanted to strike a grand bargain with Patrick McHenry, who's on his way out. So McHenry on the Republican side and Waters on the Democratic side in the House may be finalizing some of these details about stablecoins where we very well might see a stablecoin bill by the end of the year or perhaps by early 2025. Definitely look forward to hearing what my fellow panelists have to say about that. Regarding the election, just real quick: to me, there are three major events in crypto that have impacted the United States. The first one was in 2019, when Facebook introduced Libra, or now Meta, and that woke up members of Congress to what this was all about. They realized that this global stablecoin concept that a social media company wanted to introduce around the world as their own way of developing money really scared them and brought the members of Congress's attention to what this was all about.

In 2021, you saw the crypto tax proposal on how exchanges like Coinbase might report their crypto taxes to the IRS. And that woke up the lobbying element of crypto, where before it was like four or five of us, and then it was like 400 or 500 of us involved in the lobbying activities of crypto that you see now. So 2021 really broadened within Washington, D.C., to make crypto and stablecoins their own event. I call what we're seeing right now, in 2024, a very interesting point in time—sort of the awakening across the entire United States about crypto. You have voters who understand crypto, polling shows over half of voters want their leaders to have some kind of informed opinion about crypto, and I think that just represents how far we've come. Every voter counts.

I'll say this in honor of Amanda Wick. We have somebody from one of the parties doing an airdrop of a million dollars a day in my home state of Pennsylvania, trying to get you to vote one way or another. What that shows, of course, with what Elon Musk is doing on X, is that these votes are critical. We have these critical battleground states, and every vote matters. So when you have an issue like cryptocurrency, that's new and burgeoning, and everyone’s still figuring out how to position themselves regarding it, it could be a real game-changer and the X factor in this election. So that's why you see the industry, policymakers, and people in both the Trump and Harris campaigns spending time investigating how crypto might affect voter behavior. Now, there are many other issues that may be more important in the election than crypto, but the fact that it's part of the conversation and that it’s such a razor-thin election means that we've really awakened the entire country, including voters and policymakers, to understand what crypto is and to care about the positions of policymakers and potential officeholders on crypto.

So we want to know what the discussion points for regulatory uncertainty are from global perspectives. Sure, yeah. My name is Josh Besco, and I'm the head of compliance strategy at Talos. If you're not familiar with Talos, it's like an infrastructure play. It's a service provider where you can outsource if you're an institutional trader, hedge fund, asset manager, or

custodian—a way to connect all different APIs, access all your liquidity on one screen. Simply log into Coinbase, Binance, and everything else. So if you're an advanced trader, this is how you access the market to sell. Hedge funds access the market; they don't log into Coinbase. They access all liquidity, find the best price, allocate it, and money moves quickly. That's what Talos does.

Prior to Talos, I spent 16 years at AQR Capital Management, a large quantitative hedge fund, mutual funds usage provider, and then also at JP Morgan in the risk department. The past two years at Talos have been interesting. My job is a bit of everything, a bit of going around the globe, talking to different regulators, a bit of trying to kind of explain what we do to our customers and how they can talk to regulators about that—this is a good thing where you're kind of outsourcing to a safe company, getting better execution for your end client, etc., and developing products that are compliant with things like MiCA or any sort of US rule, if and when that is.

Especially this past year, traveling around the world and speaking with regulators, they kind of assume nothing's happening. They assume it's a mess, let's be honest, around here. That's just the way things go. Everybody's very excited about the ETF. They're very interested in the US election. But I think they don't have the same level of visibility, and they can't name the bill or the stablecoin bill. And, you know, they look in their own jurisdiction, and many of them, most of them—the large majority of them—already have rules. They've had rules forever, in terms of crypto forever. You know, they've had stablecoin legislation. They've had market structure regs already. And then even Europe, you know, coming in with a larger piece like MiCA. So MiCA, I think, is much more discussed, and questions are being asked because there's actually something to ask questions about.

The US is interesting, I think, for a few of the things adding on to what Jason mentioned. There's a lot of money. It's like, what's the largest PAC or second-largest super PAC with the 200-plus million dollars? So that wakes a lot of people up. I am always amazed at how many single-issue crypto voters I meet and encounter. It's just shocking. It really is crazy that they might believe and feel differently for a variety of issues that might sway them one way or another, but crypto is what's going to win their vote. It's crazy to say, but it's just true.

One last thing I'll say on the US is that there's still an image problem. I have friends that I consider close friends, so hopefully, they would know what I do, but they assume that crypto is spending money to avoid regulation. I've never been part of an industry that wants regulation so badly. It's just like dying for regulation. They want rules in place. So it's interesting in the narrative to bring people in with experience from being regulators, from traditional finance, whatever it may be, to mature these conversations. We were talking about this a bit earlier. So I think we will continue to see that certainly in the US and internationally, a more mature conversation about regulation, and hopefully end up in a good place here.

Okay, so now that we have an overview of global perspectives and US perspectives about crypto regulations in general, I think we can finish this section, but if you have any questions around what was discussed, feel free to ask.

We would like to move on to the institutional adoption section. When we created this agenda in the working group, we discussed what "institutional adoption" means, particularly how we define "institution." This is an important question. Lesley will present about banking adoption, so she will give us a perspective on banking as an institution.

So, everybody can see that. From the organizers' perspective, we thought about how to make these regulatory challenges and uncertainties facing the digital assets industry tangible. I wanted to give an example of the banking challenges crypto firms face in the United States.

I'll start with a brief introduction of myself. My name is Leslie Chapkin. I'm the head of global policy at Paxos. Paxos is a regulated tokenization and blockchain infrastructure platform, providing technology to tokenize, trade, custody, and settle assets for enterprise clients. We power the stablecoin solution for Stripe and also work with PayPal, MasterCard, and MercadoLibre, among other institutional players. Previously, I worked for the Stellar Development Foundation, JP Morgan, and spent over a decade at the Treasury Department.

While crypto is designed to operate outside the traditional financial sector, the reality is that the industry still relies on banking partners for essential services like safeguarding customer assets and providing daily banking rails. These partnerships help crypto firms operate more securely and efficiently while staying compliant with global regulations. Encouraging this activity within banks also allows US regulators to monitor digital assets, preventing the movement of fiat into unregulated areas, which could increase systemic risk.

Before diving into the main points, I acknowledge that regulatory barriers are only part of the issue. Operational and technical challenges still exist, as many banks lack the infrastructure or expertise to handle digital assets. The costs to upgrade systems can also be prohibitive. However, these are obstacles the private sector can address over the medium term. The regulatory challenges, on the other hand, require a shift in US regulators' mindset regarding digital assets.

Guidance from US policymakers has created a chilling effect on banks' willingness or ability to serve the digital assets industry. For example, a joint statement from the Federal Reserve, OCC, and FDIC last year highlighted risks with crypto, especially the use of open public blockchain networks. This discouraged banks from serving the industry. In a policy statement, the Fed presumptively prohibited state member banks from holding crypto as principal, deeming it inconsistent with safe and sound banking practices.

This stance isn't just limited to banking regulators. The White House also released a roadmap to mitigate crypto risks, characterizing digital assets as a financial system threat. This guidance

preceded the closures of Silvergate Bank and Signature Bank, major crypto-friendly institutions. With these closures, the number of crypto-friendly banks in the US has drastically declined. Customers Bank, a remaining crypto-friendly bank, recently faced an enforcement action requiring 30 days' notice to the Fed before any new crypto-related activities.

These actions remain a significant barrier to securing and maintaining banking partnerships in the United States, which could limit the industry's growth and pose risks to both the industry and consumers.

Here are some key risks: First, major banks are cautious about their regulators' views on digital assets. This has forced crypto firms, particularly smaller ones, to rely on small and medium-sized banks, which increases concentration risk, especially after the collapse of Silvergate and Signature. Second, some firms are looking overseas for banking partners, potentially complicating international money transfers for US customers. This also affects the stablecoin market, which is dominated by US dollar stablecoins. US regulators would benefit from encouraging these assets to be held domestically.

To address these barriers, I believe Congress should require federal bank regulators to clarify their concerns regarding digital assets and open the rulemaking process for public input. Additionally, as a stablecoin issuer, Paxos views the digital asset sector as a systemically important lender to the US government. Stablecoin issuers invest heavily in US Treasury debt, which should be viewed as a valuable source of funding. US banks should be encouraged to offer safekeeping services for these assets, supporting financial stability and innovation.

It's crucial to recognize that the industry is not a monolith. Some firms adhere to rigorous licensing and oversight, which helps address safety concerns from the outset.

So, that concludes my presentation, and I think it highlights both the regulatory environment and the banking challenges hindering institutional adoption of this technology.

We would like to discuss two main points: the banking sector's adoption of crypto assets and the issue of banks holding reserves for stablecoins. We'd like to delve into the regulations and global perspectives around these topics.

Let's open up the discussion. I'll hand it off to discuss SAB 121 and how it has limited banks from entering the custody business. Later, we can address why institutional investors value bank custody for crypto assets.

From a regulatory standpoint, there's a lingering sentiment that digital assets could trigger another financial crisis. Current regulators, particularly those shaped by the 2008–2009 crisis, often view digital assets with suspicion. However, unlike the housing market, digital assets function in a different context and present distinct risks. Regulators need to better understand this

technology. One major issue is that they cannot hold or directly interact with crypto assets, which hinders their understanding.

Additionally, the approach some regulators have taken has led to what's called "Operation Choke Point 2.0," where they've essentially discouraged banks from working with crypto companies. While Operation Choke Point was a covert effort, this time it's overt, with agencies and even the White House signaling that crypto might be too risky for the banking system. This creates a challenging environment for banks, as they fear regulatory pushback.

Interestingly, Bank of New York Mellon recently got approval under SAB 121 to hold Bitcoin and Ethereum ETFs. While this shows progress, it was a closed-door exception and doesn't provide a transparent or replicable model for other banks. The lack of published examination guidelines for crypto or stablecoin custody is also problematic, as it creates uncertainty about regulatory expectations.

This approach limits innovation and growth, as all banks and stakeholders should have equal opportunities to explore and safely hold crypto assets. Allowing more banks to hold these assets would give regulators the insight they need to develop appropriate oversight.

And to add to this point about Operation Chokepoint 2.0, the efforts here are not hidden; everything is quite public, with statements explicitly cautioning against crypto in banking. Despite this, there is still strong interest among agency staff to understand digital assets and develop a regulatory framework that allows for innovation.

On the international front, there's growing regulation around digital assets in places like the UAE, Singapore, Japan, and the EU. These regimes, while advancing, lack alignment, creating silos in how crypto is managed. The United States could play a more prominent role in setting standards and providing guidance for stablecoins and crypto, filling a current void in global leadership.

Regarding SAB 121, it prevents banks from holding crypto without sufficient additional assets to back them, viewing crypto as high-risk. This requirement complicates bank custody for crypto, making it difficult for institutions to offer these services. Asset managers prefer not to custody their own assets and usually rely on third-party custodians for security and regulatory compliance. This lack of custody options restricts larger asset managers, who would otherwise invest more heavily in crypto.

Institutions like BlackRock and other asset managers want to diversify by adding assets like Bitcoin, which have different risk profiles than traditional stocks and bonds. This diversification improves portfolio performance, making Bitcoin and other assets attractive for institutional investors. However, the lack of trusted custodians hampers their ability to invest as they would in

other asset classes. Thus, regulatory clarity and safe custody options are crucial for institutional investment in crypto assets.

Your assets, that's where if you have a pension or a 401k, that's where you would want your money. The other thing, too, and you might think about this with different countries, is it's actually an oddity that the SEC made such a pronouncement about how to hold crypto custody because they're not really a direct banking regulator. You know, if I were to talk to colleagues at JP Morgan Chase or the Treasury, and I said, "Well, the SEC said to do it this way," they'd look at me like, "Why are you following this?" This shows how new this all is because, in the US, different regulators oversee different parts of the market. The SEC, for better or worse, is trying to position itself as the main regulator for digital assets, exerting its authority by specifying how SEC companies must account for crypto assets.

Even the Federal Reserve admitted to Congress that the SEC's guidance confused them. Typically, it's the banking regulators, and banks are traditionally the most trusted entities for custodianship, considered qualified custodians. What we need, hopefully illustrated by the Bank of New York Mellon example, is a regulatory shift to address the straightforward issue of how crypto assets are held. Billions of dollars in stablecoins and crypto asset custody are waiting to be unlocked if the banking regulators can clarify their stance. There's a bit of a chicken-and-egg problem, as some regulators worry about potential risks if the market grows significantly. The industry needs to do a better job explaining the economic benefits of crypto and how risks can be managed effectively.

For instance, I'd rather my stablecoins go through a Know Your Customer (KYC) process than traditional cash transactions. The crypto industry has been on the defensive for the past eight or nine years. We also need to go on the offensive by highlighting the benefits, showing how this can actually support the banking system. Most people agree that digital assets are the future, but regulators, as always, are cautious because it represents a significant change, keeping them in reactive mode.

There's a need for proof-of-reserve rules, companies like Circle, and assurance that the ecosystem is safe. When we show that the ecosystem operates with responsibility and concern for security, trust increases. We're not here to YOLO and "go to the moon"; we genuinely believe this is the right path forward.

I'd add that the absence of proactive policymaker engagement in this space is leading to opaque, offshore US dollar stablecoin issuers. This is not the transparency we want in the US. Creating frameworks to encourage the development of a US-based US dollar stablecoin industry is crucial. Transparency, proper reserve management, and public attestation should be the foundations for growth in this sector. Right now, issuers with little to no regulatory oversight dominate the market, which is not ideal.

The industry is still developing, and smaller crypto companies often respond to hacks by quickly announcing on platforms like X, "We've been hacked, but your funds are safe." Regulators see this, and it doesn't inspire confidence. We need to grow up as an industry and have more seasoned professionals at the table. Regulators often see themselves as partners, like consultants, wanting businesses to operate safely and within the law. However, when offshore entities act opaquely without standardized practices for cybersecurity, it makes regulators wary. Establishing standard practices, such as what to do in the event of a hack, will be essential.

Crypto is not alone in facing risks of money laundering violations. We recently saw TD Bank in the news for a significant money laundering violation. So, while the industry is still in its early days and has some growing pains, we're not alone in facing regulatory challenges.

If we want to avoid a shadow banking system, proper regulation is necessary. The real systemic risk comes from keeping crypto out of the regulated banking system. For example, the concern around contagion was flipped on its head with the collapse of Silicon Valley Bank, showing that the risk isn't in crypto seeping into traditional finance but potentially the other way around.

On the topic of cybersecurity standardization, there will be a session about it tomorrow as part of a larger effort to create cybersecurity standards and alignment within the crypto industry. BGIN is actively working on this.

We also have a question about banking adoption and stablecoin holding by banks. A great question about whether US regulations will allow changes or if we need the BIS rules to adapt first. There should be changes in how the BIS views crypto assets, particularly anything on a permissionless network. Stablecoins, even fully backed ones, are considered high-risk with higher requirements. This is an essential angle to consider, and the US would benefit from aligning with other regulators on this. Country-by-country frameworks don't align perfectly, and harmonization is necessary for practical regulations around stablecoins.

The BIS recently allowed banks to hold a small portion of assets in crypto, which doesn't automatically translate to US regulatory changes. While the BIS focuses on broad issues like stablecoin versus CBDCs, the core question remains on stablecoin regulation within the monetary policy framework.

It's practical to start with a small percentage to build infrastructure gradually. Unfortunately, the US doesn't typically have digital sandboxes, unlike other countries, which would allow for these incremental steps.

Any questions from the audience?

I want to comment on the recent security token issued by Black Rock on Ethereum. Stablecoin issuing companies are considering supporting security tokens, which could improve

transparency. Security tokens issued only through private offerings may lack the visibility that an on-chain system can provide.

Regarding on-chain collateral, US policymakers aren't fully ready for this shift yet. However, the transparency benefits make it a conversation worth starting. It aligns with the idea of tokenizing real-world assets, which seems more achievable in the short term. As we look to 2025, tokenization of securities and assets is likely to become a significant topic, and understanding how to regulate tokenized assets will become more relevant. There's still a need for both visibility on permissionless blockchains and privacy that traditional banking also values.

This has been a very informative discussion.

One thing, coming from the industry perspective—and I'm kind of the inverse of everyone here—I ran a crypto asset firm and then discovered all the regulations and conversations. I even testified to the Senate around debanking. That's how I found myself in these rooms. One thing that seems glaringly obvious to me but might not be to others is that there's a middle ground in regulation between the decentralized, permissionless side and the centralized side. This middle ground could be bridges. I think there's an issue with terminology and understanding in the crypto industry. There are centralized bridges like Talos, which is more of a bridge between multiple liquidity venues. And there are decentralized bridges as well. These bridges act as trust nodes or messaging protocols between different blockchains.

I believe regulators could create specific rules, or at least cybersecurity regulations, around asset holdings, collateralization requirements for bridges, and defining which assets can bridge between certain chains. One of the most successful examples is Circle's USDC stablecoin bridge to Solana. A lot of people wanted to move their assets from Ethereum to Solana to trade, and it's gotten a lot of use. I trust that bridge. I don't use third-party bridges that are likely developed offshore. This isn't just a statement but also a suggestion that we need better-aligned terminology. Banks are like trust nodes—they're the bridges of finance globally. Maybe a bridge is just a bank, and a bank is a bridge.

It would be great if we had common terms for many things in crypto. The industry is good at creating digital versions of pre-existing concepts, but having unified terminology would be beneficial, especially for regulators who need to understand these concepts at a basic level.

The bridge issue is interesting. It reminds me of some traditional finance practices under MiCA and other jurisdictions, where material outsourcing requirements require thorough due diligence. If you're putting money into a bridge, you have an obligation to understand how it works, ensure it's safe, and regularly verify it can't be compromised. At our company, we proactively seek SOC reports and other certifications to prove we take security seriously. We try to create thoughtful policies and procedures, even if the space remains somewhat gray. Ultimately, you can either do your best with what you have or not enter the market.

Banks could indeed serve as trusted nodes, which is a solid way of viewing their role. Bridges serve as the main pathway for interoperability, and creating specific standards around bridge cybersecurity could provide real progress, as opposed to another abstract discussion on interoperability.

Interoperability is the most centralized point of the DeFi journey. If institutions serve as trusted nodes, then bridges might be the ideal place for them to operate.

Using consistent terminology is essential. There's much that everyone should agree on, and industry groups should strive to create unified standards. Regulators could do better in this area. When I worked with CFTC working groups, we sometimes aimed for sweeping statements, but often, we just need clear definitions.

We saw the Dutch regulator recently release a simple PDF with rules under MiCA, including straightforward "do not" examples, like pump-and-dump warnings. It might seem basic, but we need these kinds of starting points. Clear guidance can help design technology to detect issues like market manipulation. Various technology providers are ready to help, and establishing common definitions, policies, and standards would be invaluable.

In the crypto space, we're sometimes the "Rodney Dangerfield"—we get little respect compared to fields like AI, where regulators are encouraged to experiment and engage. Crypto's foundational concepts are changing how we view money and technology. We have blockchain governance, digital forms of money, bridges, and stablecoins, which could upgrade the financial system. But Bitcoin's presence alone often causes a full stop in regulatory discussions. When we mention stablecoins or Bitcoin, it becomes overwhelming for some regulators.

My approach is to separate these discussions. I'm not here to debate the future of money; I'm here to explain how the technology works and why institutions see value in it. It's crucial to make these distinctions and keep the conversation grounded in cybersecurity and functionality, avoiding debates that make regulators defensive.

Terminology matters, especially with a multi-stakeholder audience. In many BGIN discussions, people think they're talking about the same thing, but different definitions lead to confusion. Bridges, being a centralized point in blockchain space, come with inherent risks. However, many regulators lack a deep technical understanding, making it challenging to address these risks. I think this is an excellent transition to the next session.

Actually, so we have this agenda. So we would like to talk about the new technology related businesses. So while, you know, Mitchell mentioned it's like centralized bridges, regulators need to have a good understanding of what are the risks, what are the regulatory risks, what are the risks from regulatory perspectives of these technologies. What we would like to do in the beginning is to have a multi-stakeholder discussion around new technologies, even though it's definitely changing in a few years or 10 years or 15 years. We have to catch up with all of the latest trends of technology to identify the regulatory risks.

And I think we can create a document to identify the risks of these technologies from regulatory perspectives, I believe.

So we would like to make a list of technologies that regulators and people may not fully understand. I think one example of new technology is centralized bridges, as Michell said, and also the Lightning Network is one of the new technologies we discussed a lot. We had a conversation with Bitcoin developers about the risks of this new technology in Tokyo, Japan. I don't want to dive into the details of what we discussed, but I'd like to give you a brief introduction of how we discussed that matter.

This is an overview of how the Lightning Network works. Does everyone know what the Lightning Network is? It's a scaling solution for Bitcoin, called Layer 2, that aims to provide scalability and reduce gas fees for transactions on Bitcoin. When users want to transact, they use a multi-sig contract, which means the keys of user one and user two are involved.

Once they lock the assets of user one and user two, they create a channel on the Lightning Network and use a Lightning Node, involving the nodes of user one and user two, to exchange commitments for future transactions on the Bitcoin blockchain. When they want to finish, they can make several transactions on the Lightning Network. To record the final state of all these transactions, they submit the final commitment to the Bitcoin blockchain. After that, user one and user two can unlock their assets from the multi-sig contract based on this final state.

If user one wants to transact with a third party, user three, without a direct connection, they have to use an intermediary node, like user two's node. Through intermediary nodes, user one can transact with user three, but there is a privacy problem, as user one can see all transactions between only user one and user two, while user three cannot see transactions between user one and user two due to privacy-enhancing mechanisms on the Lightning Network.

Once user one submits the final state of the transactions, it's not an immediate process to unlock assets on Bitcoin's main chain, as there is an approval waiting period. If user two is offline during this period, user one can still unlock assets, even if the transaction submission was incorrect, which poses a risk. To mitigate this, there is a monitoring person, known as a watchtower, who monitors transactions submitted from the Lightning Network to Bitcoin's main chain.

This is a brief overview of how the Lightning Network works. We identified the regulatory touchpoints of the Lightning Network, like user one's node, which operates behind the user's wallet. A self-custodial wallet requires private and public keys, and for transactions on the Lightning Network, a private key for the Lightning Network node. However, many self-custodial wallets rely on third parties to operate the Lightning node. This reliance introduces risks because a third party could initiate unauthorized transactions.

Operating your own node means you manage the private key and are the only one able to make transactions. But if relying on a third party, your assets could potentially be stolen. So, the user's node in the Lightning Network is a regulatory consideration, as well as the role of the monitor. There is a risk of malicious activity, such as asset seizure, which is why this role also requires regulatory consideration.

The intermediary node, although it mostly follows transactions directed by the user, does pose potential risks. We identified these regulatory touchpoints and created meeting records around them.

So, there are three primary regulatory touchpoints for the Lightning Network: self-custodial wallets, monitoring roles, and intermediary nodes. We reviewed regulatory factors like security and AML compliance. For intermediary nodes, we analyzed the risk of asset seizure and AML measures. Additionally, crypto exchanges that support the Lightning Network raise questions. More exchanges are integrating the Lightning Network for Bitcoin exchanges, which is another point of regulatory interest, as is the role of watchtowers.

We don't have time to cover everything, but it's essential to create a document to help regulators understand the regulatory risks of these new technologies.

I'm considering applying this approach to other emerging technologies that are not well-understood.

For our next discussion, I'd like to explore what other new technologies people tend not to understand well and create a list of them. For example, the Lightning Network is a Layer 2 scaling solution, but there are various Layer 2 technologies for Bitcoin, Ethereum, and other blockchains, which we need to understand thoroughly.

We also need to discuss staking as an investment. With recent concepts like liquid staking and restaking, it's become more complex. We should explore the regulatory risks associated with these as well as cross-chain protocols, as Mitchell mentioned, which also carry risks. DeFi, too, though broad, introduces AML and other risks related to AMMs and lending protocols. We should analyze these from a regulatory perspective.

I’ve listed what I came up with, but is there anything you think is missing from this list, or that people in regulatory roles may not discuss often but should be aware of? This is one of the main questions I wanted to ask you all.

Thank you. Anyone?

Well, this is a great list to start. I would say maybe also consider adding yield varying stablecoins because there has been such an explosion of them recently and they pose sort of unique regulatory challenges. So that might be worth adding to the list.

Okay.

Yeah, I think this is a great list. It's a great start.

I think I was going to say even just stablecoins, you know, as far as something that you can roll out and work with industry on to different regulatory bodies so they understand how it works. Don't assume this is called a stable coin that they all understand it.

There's lots of intricacies in what you saw with that layer two diagram I know stands between us and happy hour.

As far as the complexity of how stablecoins work. But, you know, it's important to remember too with like looking at stablecoins is speaking sort of the regulatory language behind it. So if you're going to look at something like stablecoins, you want to look at proof of reserves, which is this major term really important and something that is probably a lot more manageable than say like someone who would set up a lightning node, which can be set up by anyone.

And also want to keep in mind what the regulators are thinking about and meet them where they're at. So the unit may not be able to move off of level one to level two, right? It's very advanced. And you actually could see it that one of the people that was vetted for the OCC here in the US, I can't remember her name. She had put a paper out where she actually criticized the Lightning Network because it was taking it off the main chain, off of Bitcoin. And so these transactions were happening in her mind, almost like off balance sheet transactions and made this connection to that in the financial crisis, which I didn't think that was correct.

But it's important to understand, right, when you look at you're moving something from a concept, you're still trying to get them to understand and explain the transparency of the layer one. And they say, oh, but there's this problem with layer one. It's not moving fast enough, so we have to go to layer two.

You know, what does that look like? And I think her point in that one paper was, you know, maybe all the layer two nodes need to have money services licenses because of what they're involved with, press and be responsible for things like OFAC and fit and BSA AML.

Again, this is not rehearsed at all, but I would rather have like a paxos or a circle or a major company at the table looking at that than say somebody in their basement that just opens up a lightning node. So these are considerations you want to think about the way the regulator is looking at it too. And, you know, happy to help as you develop these things. But I think just, yeah, just stable coins. A great one.

Yeah, I think it's a good start, Liz. I would say pictures help a lot a lot like flow flow diagrams, pictures can go a long way add colors again colors always work well.

And if you can compare them to the things that they know like current products in financial industry or even something else, a flow they can understand really will be helpful.

Very quick story. I spent probably six years of my life. There was a rule around use of derivatives for mutual funds that like would have shut down a lot of the funds we had, which we felt were very safe funds, but they used a lot of leverage on very low volatility thing. So if you lever a bond it's not as risky as levering, you know, emerging market fund or something, right? That concept was understood the role was not well drafted long story short we again, six years later. But the reason I bring this up is we spent a lot of time building an appendix with with charts and colors and it's explaining kind of the notion of risk again which you know feels like something you shouldn't have to explain, but we did and it the multiple SEC regulators pointed that out as like the thing that kind of helped them understand this issue and why we were so upset and thought it was bad regulation.

So all the words in the world sometimes don't add up to one really well crafted chart or picture that anybody could understand.

And we can, I mean, if your laws can help with those sorts of things as well but i think if the end solution is even just a float, I would love to see a float drop for these, right? that you can kind of present over and over to a regulator.

I guess I would just offer one overriding theme that I see running through most of this stuff here. With new technologies there's particularly in this space, I think a bit of a conflict in that most of these regulatory regimes we're talking about presume the existence of an intermediary.

A regulated intermediary, a touch point for the regulatory system in private sector where government can monitor something. I'm a tax lawyer, so I operate in this tax space where we've seen a lot of attempts to create regimes that presume the existence of any intermediary. And a lot of the stuff you're talking about there does not have naturally an intermediary because it's supposed to be a decentralized system allowing for peer-to-peer transactions.

So in that context, I think you have to come to grips directly with that problem.

I think the way to deal with that in many cases, not all cases, many cases, is to go back to first principles of what the regulatory team is trying to accomplish. And see if there's a different way of getting at that issue. In some cases, there actually is potentially an intermediary there because of the way the market's going to develop, that it's more centralized than you really think it is.

But in other cases, there may be technological solutions to collect the information without inventing an intermediary. Because right now, the default is going to be regulators are going to want to write a rule that invents an intermediary where there isn't one already. We're seeing this a lot in the tax information collecting process is just say that a DeFi protocol is a broker.

It's an absurd notion, but it should be thrown out there because there has to be an intermediary or something. So I think that's a fundamental theme that you have to grapple with. I think that's what I get the core of biosols. At the quorum, I asked for this whole approach to DeFi, right, is just find somebody and then that is the intermediary, even if it's the developer, right? So I totally agree.

Yeah, I would just say that it's an excellent point and i think regulators need to think too about the risks that an intermediary poses as well right we saw with financial crisis intermediaries start to be bad actors, they can really mess things up too. So maybe there's some benefits to the system to have it be in this sort of decentralized way. It's very hard though for regulators to not have the model of okay, we sentenced the up to jail or something goes wrong or this or that. So it's going to take a lot of education and a real change because banks have been the lifeblood of intermediation, right, for a long time. So it's a real challenge, but it's a really good point.

Me. I saw it. Bookings and he has a you can come both ways. But we are running out of time.

Let me tell you. I just wanted to make a little coin on your opinion the uh i guess the weekend a multi-signature technology who holds the key and how it can be managed is very important for to consider regulation costing uh what do you think the wallet has money supply is two person by using the multi signatures and then make a uh maybe there's some Chattel game and if who's responsible for it even paying taxes and then what is the that's what it has been used as a tutors a tutor's a type of signatures it could be more difficult to think about so maybe yeah simple it's a partnership what if the two tourists? Everybody, I don't know.

Yeah, so that makes me a good point yeah bye.

I know, Rod. On that point, I brought up something there's been a lot, SEC was doing a sweep exam for asset managers that hold crypto in the US. They probably still are, but I think it's kind of slowed down a little bit. But to the point of like, sometimes when it's gray, you should kind of use logic and put controls in place. So you're still being thoughtful. There were some companies that were even, they would have multiple shards and they would have, you know, maybe it was

even the two founders of the company, but they would create procedures where like they couldn't be on the same plane right or like you know like but i just love that.

That effort, right? I think that's something that's kind of unique to crypto right where like there's still this they understand there's something wrong they understand the risk hopefully a lot more. I think all of us in this room do um and putting procedures like that in place because something else there is just kind of innovative. I thought that was cool. A little weird thing to try to control from a compliance officer. Yeah.

Can't take that bike together.

Right so yeah we have. And so thank you very much for the main discussions and discussing to speak up thank you very much.